

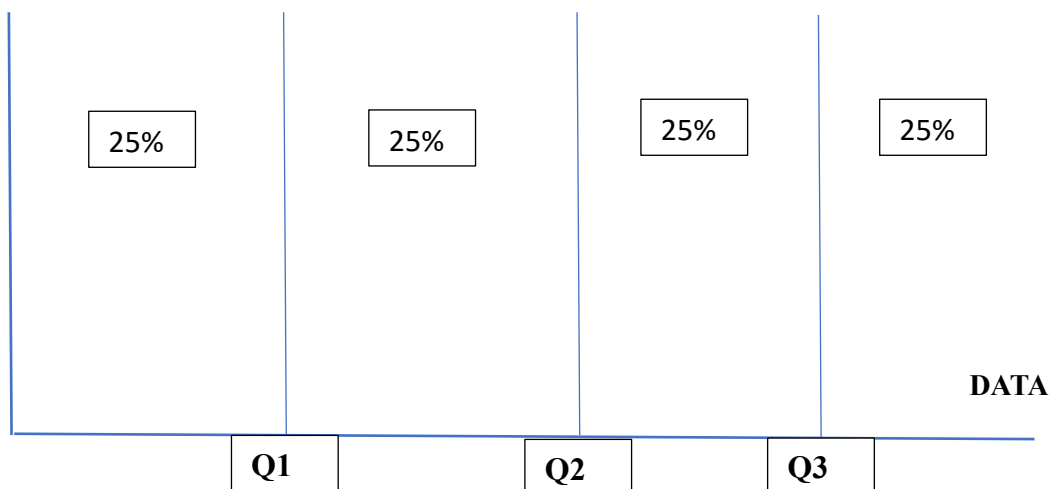
FIVE NUMBER SUMMARY USING PYTHON

- Descriptive analysis summarizes and organizes data to highlight key patterns using measures such as mean, standard deviation, and range.
- It provides a clear snapshot of the data's *central tendency, variability, and overall distribution* without making predictions or inferences.
- This analysis also helps us identify extreme data points or values that differ significantly from the rest of the dataset.
- Statistically, these are called *Outliers*.
- Such analysis can be visualized using a graphical representation called a *Box and Whisker Plot*, or presented numerically and called as "*Five Number Summary*".

Five Number summary has following components:

- Mean: - represent the average of the dataset.
- Min
- Lower Quartile (Q1): - 25% or 1st Quartile
- Median (Q2): - 50% or 2nd Quartile
- Upper Quartile(Q3): - 75%, 3rd Quartile.
- Max

Quartiles are specific percentiles that hold greater importance because they divide the data distribution into equal parts.



Some examples where this analysis is used include:

- Monitoring business trends.
- Understanding user engagement and content reach on social media.
- Summarizing patient health statistics.
- Analyzing student performance data.

Let's see how we can perform five number summary analysis in Python. There are various approaches or techniques one may use depending on their needs.

Consider the following data representing the number of passengers on specific flights between Delhi and Hyderabad over the last 33 days: {128, 121, 134, 136, 136, 118, 123, 109, 120, 116, 125, 128, 121, 129, 130, 127, 119, 114, 134, 110, 136, 134, 125, 128, 123, 128, 133, 132, 136, 134, 129, 132, 145, 178}

Analyze the descriptive statistics to gain a basic understanding of the data.

```
#Importing Libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Unsupported Cell Type. Double-Click to inspect/edit the content.

```
#Ignore Warnings
import warnings
warnings.filterwarnings('ignore')
```

Used in Python to suppress warning messages that would otherwise appear during code execution.

```
Data=[128,121,134,136,136,118,123,109,120,116,125,128,121,129,130,127,119,114,134,110,136,134,125,128,123,128,133,132,136,134,129,132,145,17
```

```
print(f'Data type of the give data set is {type(Data)}') # To understand the type of data set.
```

```
↗ Data type of the give data set is <class 'list'>.
```

```
Data = np.array(Data) #Convert the list of an array
Data
```

```
↗ array([128, 121, 134, 136, 136, 118, 123, 109, 120, 116, 125, 128, 121,
        129, 130, 127, 119, 114, 134, 110, 136, 134, 125, 128, 123, 128,
        133, 132, 136, 134, 129, 132, 145, 178])
```

```
Passengers = pd.DataFrame(Data,columns=['Number of Passengers']) # This code will help us to understand how we can create a dataframe.
```

```
Passengers.head() #To get the Five 5 sample data
```

```
↗
```

	Number of Passengers
0	128
1	121
2	134
3	136
4	136

Next steps: [Generate code with Passengers](#) [View recommended plots](#) [New interactive sheet](#)

```
print(f'\nFIVE NUMBER SUMMARY\n')
print(f'This data has {Passengers.shape[0]} rows and {Passengers.shape[1]} columns.')
print("Mean : ",round(Passengers['Number of Passengers'].mean(),2))
print("Min : ",round(Passengers['Number of Passengers'].min(),2))
print("25% : ",round(Passengers['Number of Passengers'].quantile(0.25),2))
print("50% : ",round(Passengers['Number of Passengers'].quantile(0.50),2))
print("75% : ",round(Passengers['Number of Passengers'].quantile(0.75),2))
print("Max : ",round(Passengers['Number of Passengers'].max(),2))
```

```
↗
FIVE NUMBER SUMMARY
```

```
This data has 34 rows and 1 columns.
Mean : 128.56
Min : 109
```

```

25% : 121.5
50% : 128.0
75% : 134.0
Max : 178

```

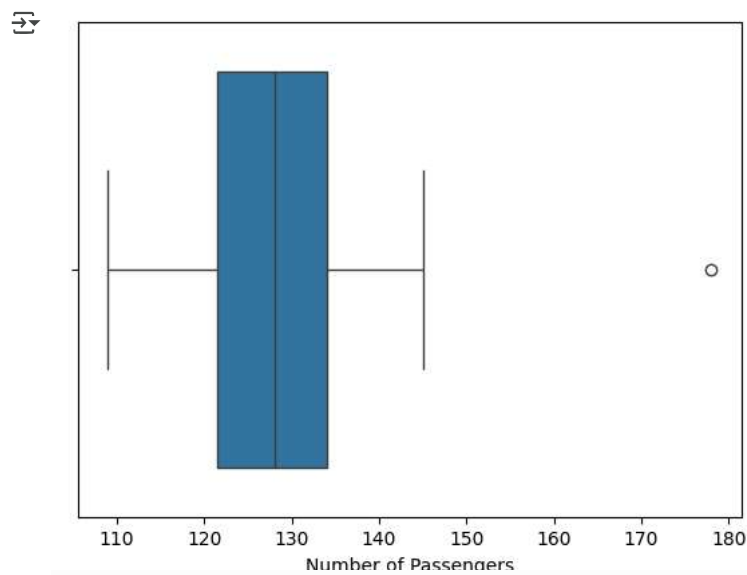
Passengers.describe() #We can use describe method to understand the five number summary

	Number of Passengers
count	34.000000
mean	128.558824
std	11.906686
min	109.000000
25%	121.500000
50%	128.000000
75%	134.000000
max	178.000000

```

sns.boxplot(Passengers['Number of Passengers'], orient='h')
plt.show()

```



```

fig=plt.figure(figsize = (6,5))
# Plot the boxplot
sns.boxplot(Passengers['Number of Passengers'], orient='h')
plt.title("Boxplot with Median and Quartiles")

# Get quartiles
q1 = Passengers['Number of Passengers'].quantile(0.25)
q2 = Passengers['Number of Passengers'].median()
q3 = Passengers['Number of Passengers'].quantile(0.75)

# Add annotations
plt.axvline(x=q1, color='orange', linestyle='--', label='Q1')
plt.axvline(x=q2, color='red', linestyle='--', label='Median (Q2)')
plt.axvline(x=q3, color='green', linestyle='--', label='Q3')
plt.legend()
plt.show()

```



Boxplot with Median and Quartiles

